

Demographic Forecasting

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Harvard University

Joint work with Federico Girosi (RAND)
with contributions from Kevin Quinn and Gregory Wawro

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 - demography & sociology
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 - economics & social security and retirement planning
 - actuarial science & insurance companies
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- Other results we needed to achieve this original goal

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- Better Bayesian priors
- forecasts and farcasts based on much more information

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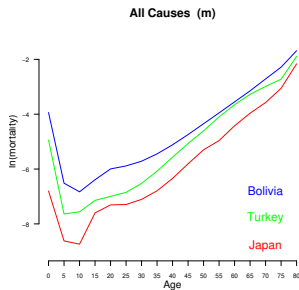
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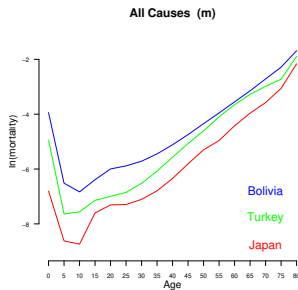
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- Most time series are very short. A majority of countries have only a few isolated annual observations; only 54 countries have at least 20 observations; Africa, AIDS, & Malaria are real problems

Existing Method 1: Parameterize the Age Profile

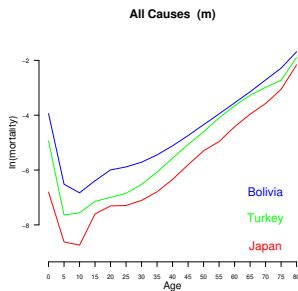


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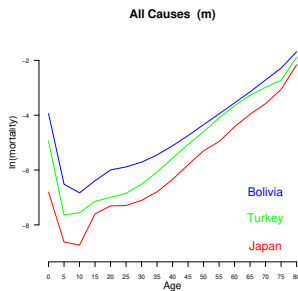
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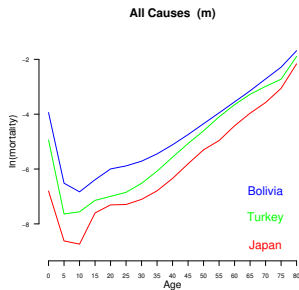
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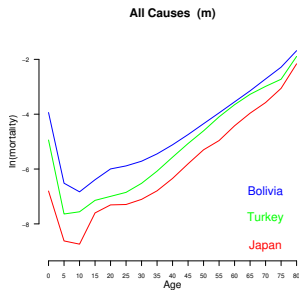
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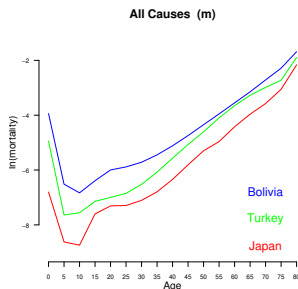
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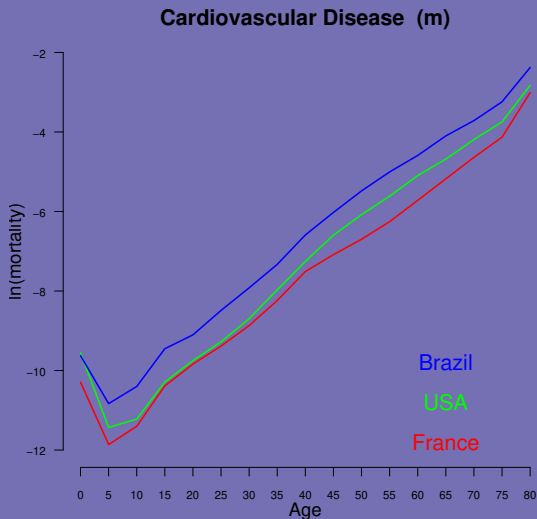
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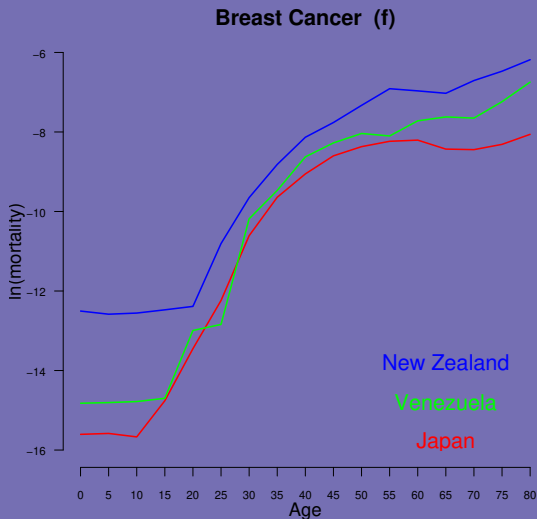


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- Dozens of more general functional forms proposed
- But does it fit anything else?

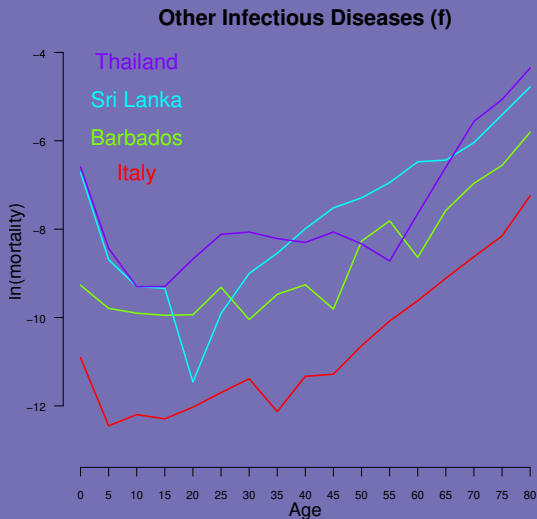
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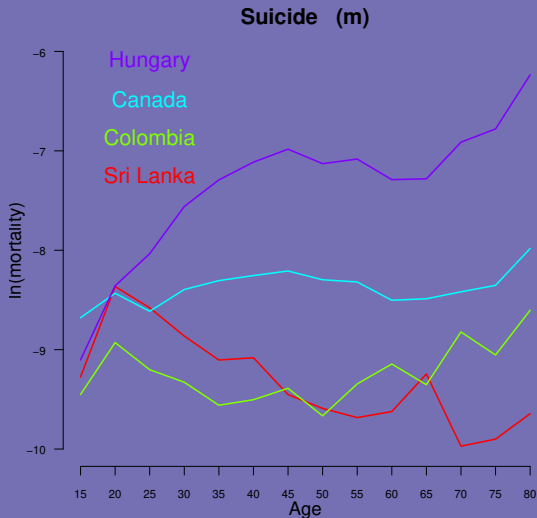
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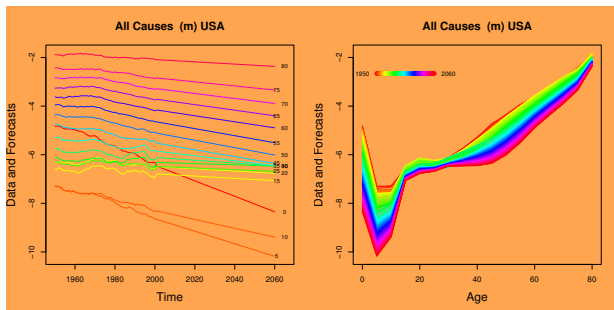
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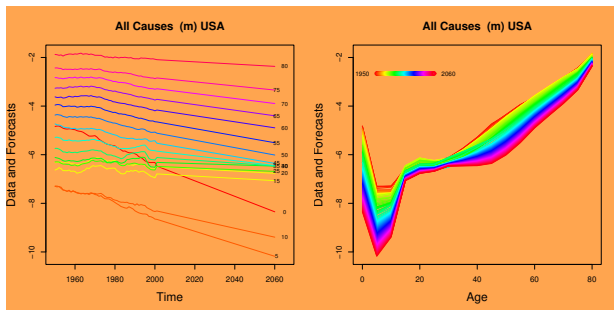
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- Ignores covariate information

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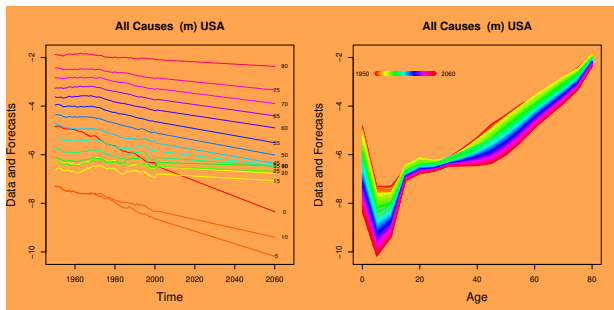


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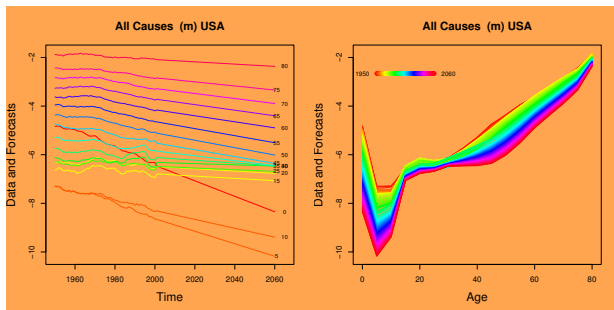
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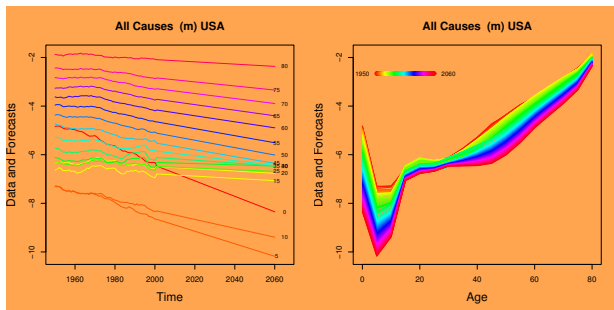
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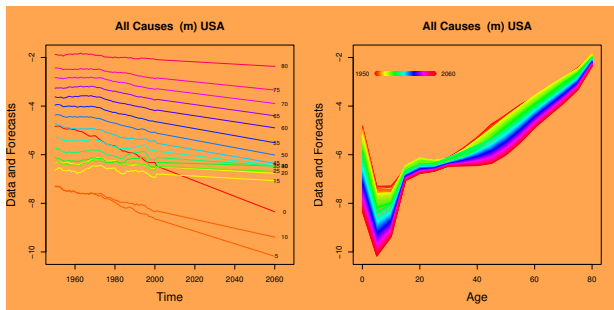
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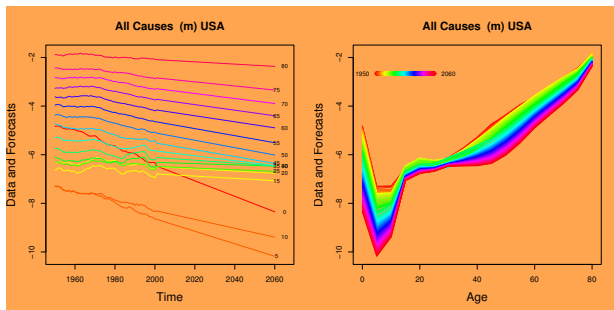
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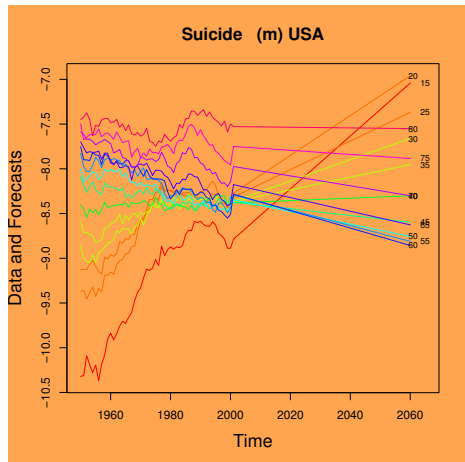
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- Does it fit elsewhere?

The same pattern?

Random Walk with Drift $\approx$ Lee-Carter $\approx$ Least Squares

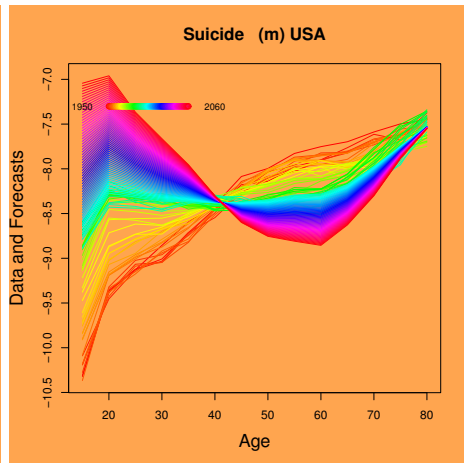
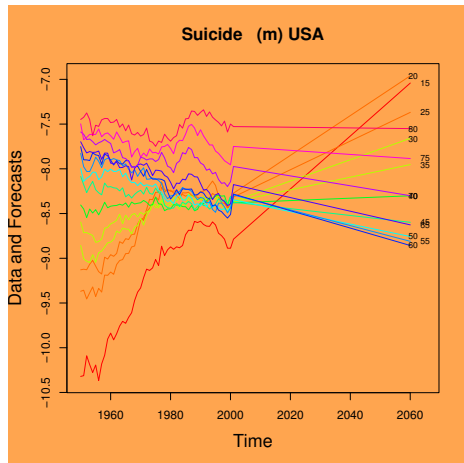
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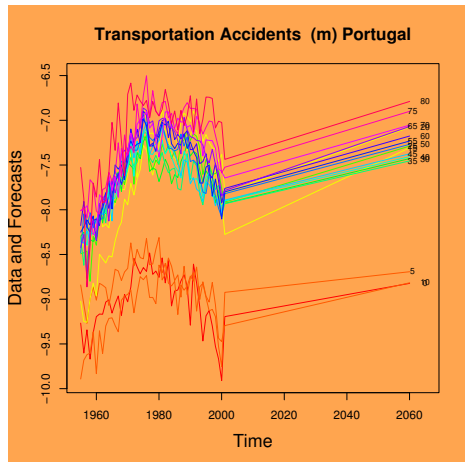


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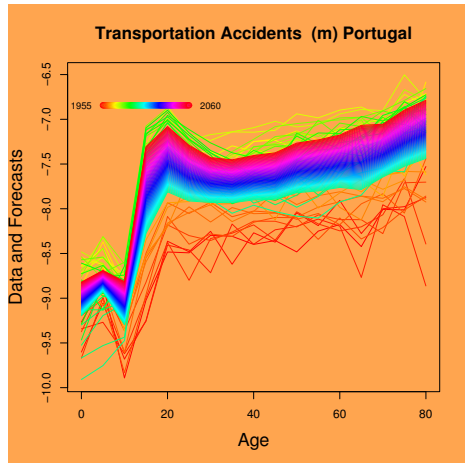
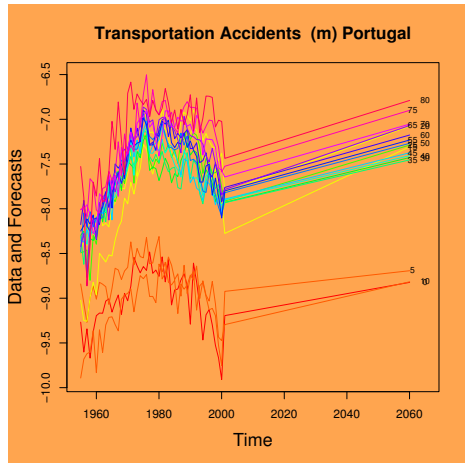
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 - same covariates required in all cross-sections

Partial Pooling via a Bayesian Hierarchical Approach

- Likelihood for equation-by-equation least squares:

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- The easy part: *easy-to-use software* to implement everything we discuss today.

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The (Problematic) Classical Bayesian Approach

Assumption: similarities among cross-sections imply similarities among coefficients (β 's).

Requirements: Comparing β_i and β_j

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Natural choice for the prior:

$$\mathcal{P}(\beta \mid \Phi) \propto \exp \left(-\frac{1}{2} \sum_{ij} s_{ij} \|\beta_i - \beta_j\|_{\Phi}^2 \right)$$

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 - 3 In the subspace, we can invert $\mu = \mathbf{Z}\beta$ as $\beta = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mu$, giving:

$$\mathcal{P}(\beta \mid \theta) \propto \exp\left(-\frac{1}{2}H[\mu, \theta]\right) = \exp\left(-\frac{1}{2}H[\mathbf{Z}\beta, \theta]\right)$$

the same prior on μ , expressed as a function of β (with constant Jacobian).

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- We start with one-dimensional $P(\mu_{cat})$, and treat it as the multidimensional $P(\beta_{ca})$, constant in all directions but $Z_{cat}\beta_{ca}$.

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- The normalization matrix Φ is unnecessary (normalization is performed by $\mathbf{Z}$, which is known)

Basic Prior: Smoothness over Age Groups

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- where W^n is a matrix uniquely determined by n and θ

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where we have defined:

$$\mathbf{C}_{aa'} \equiv \frac{1}{T} \mathbf{Z}_a' \mathbf{Z}_{a'} \quad \mathbf{Z}_a \text{ is a } T \times d_a \text{ data matrix for age group } a$$

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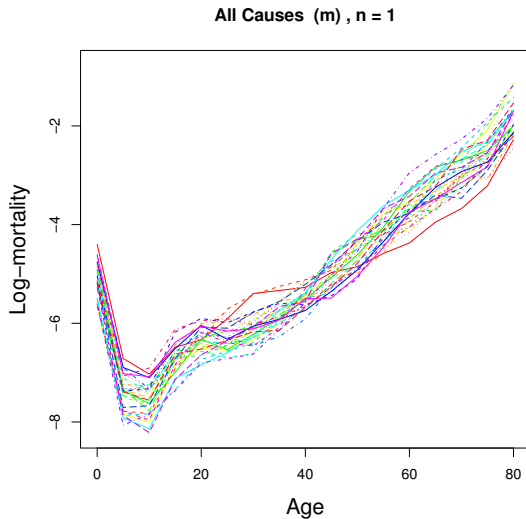
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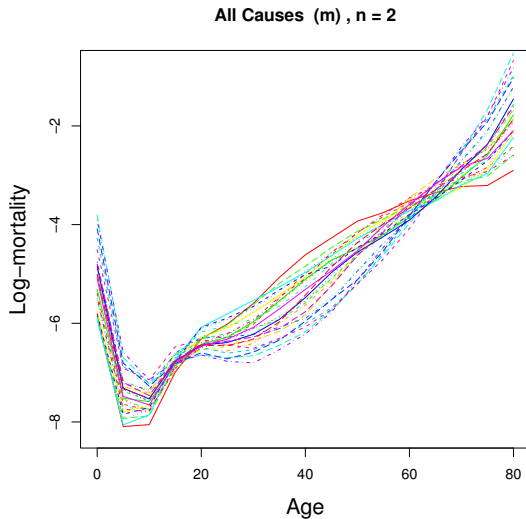
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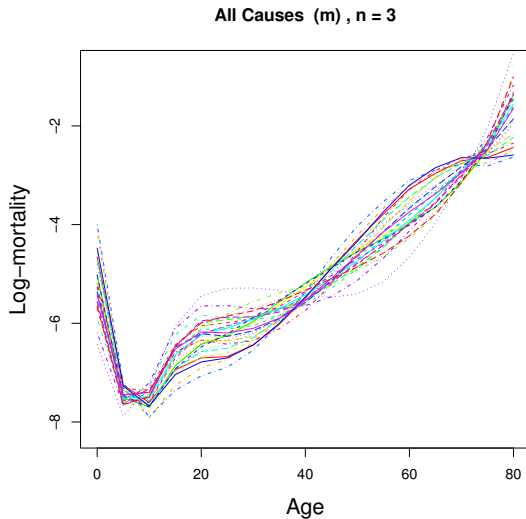
Samples From Age Prior



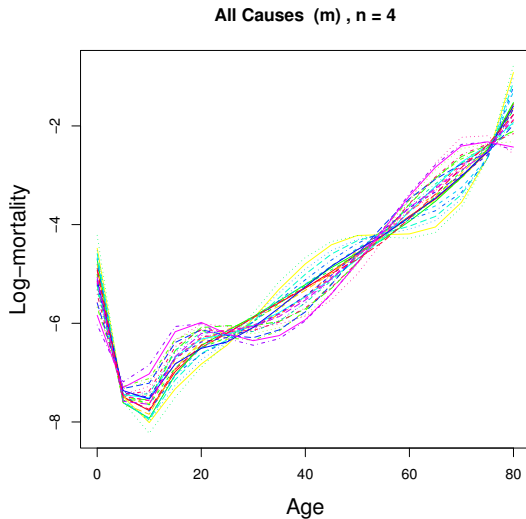
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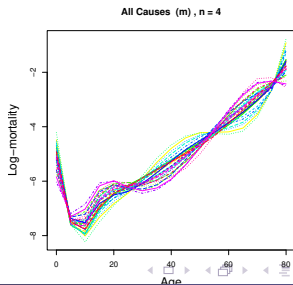
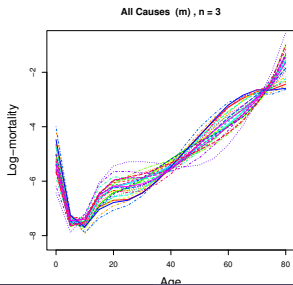
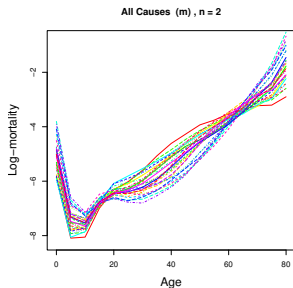
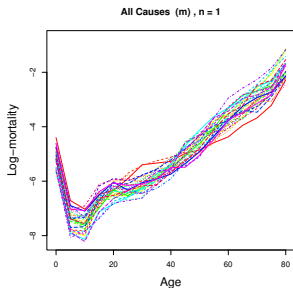
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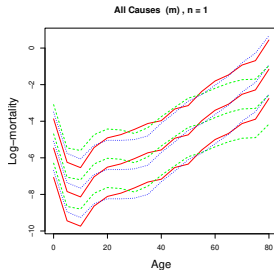
Formalizing (Prior) Indifference

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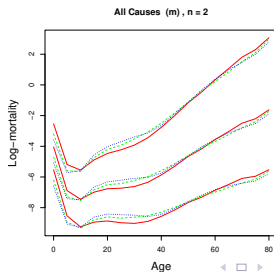
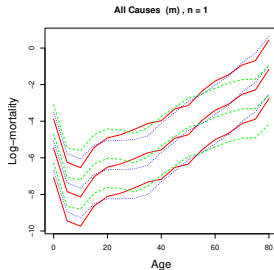
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Level and slope indifference

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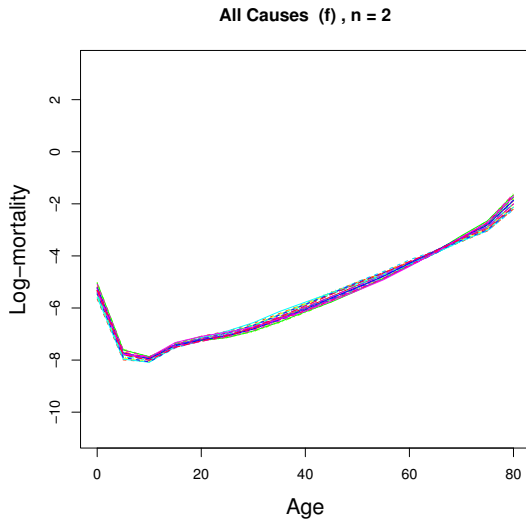
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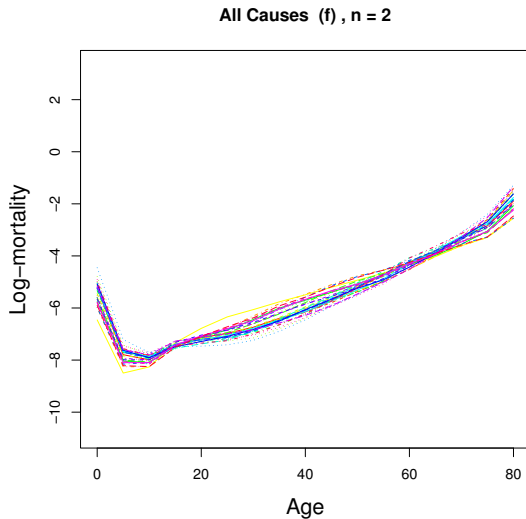
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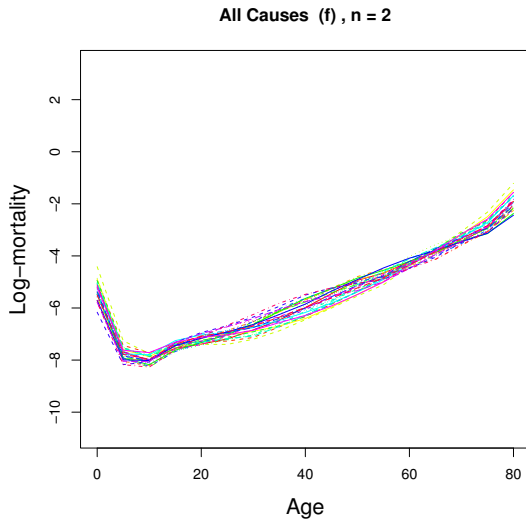
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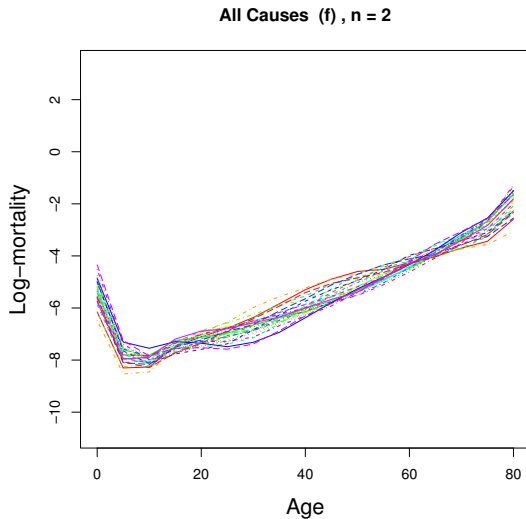
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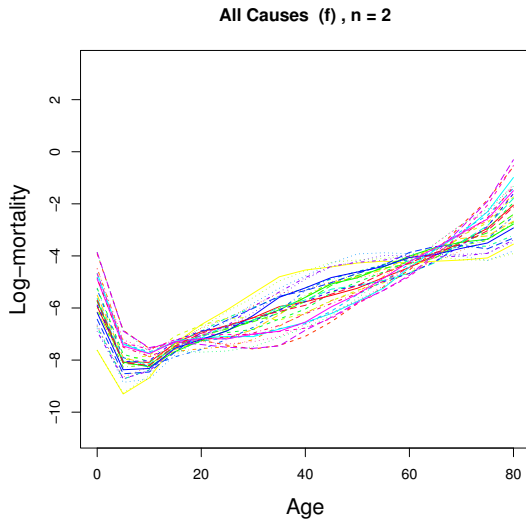
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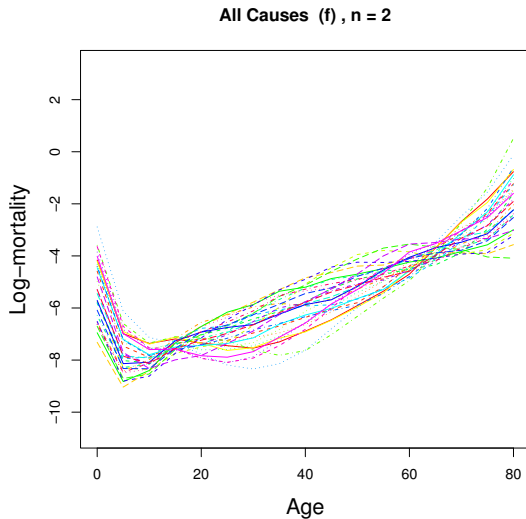
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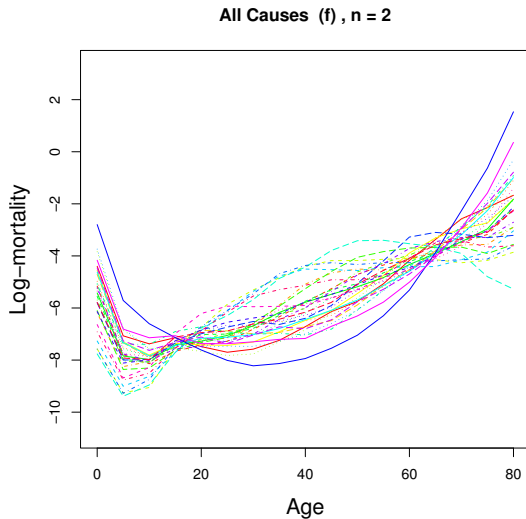
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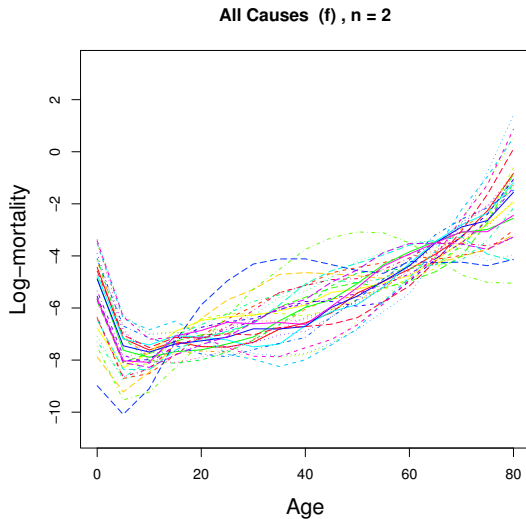
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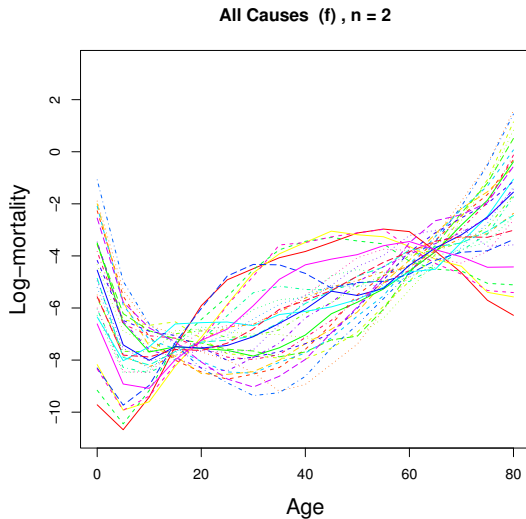
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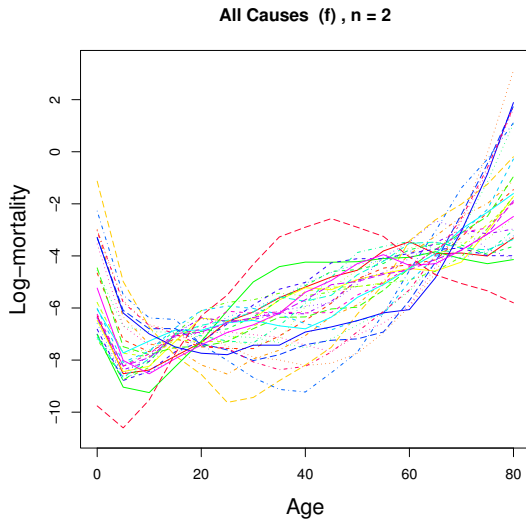
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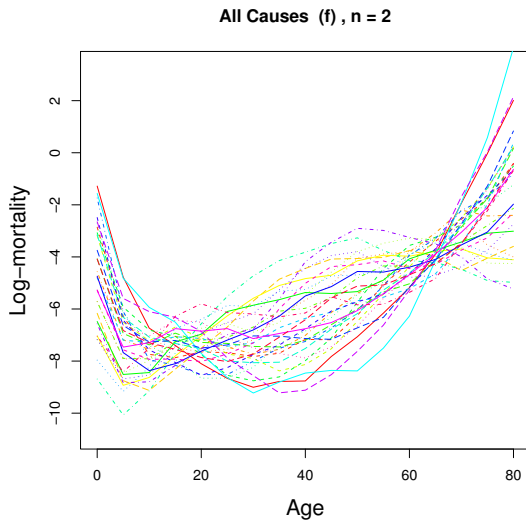
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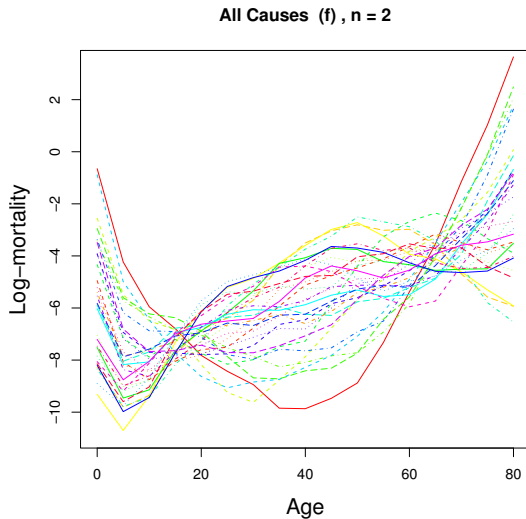
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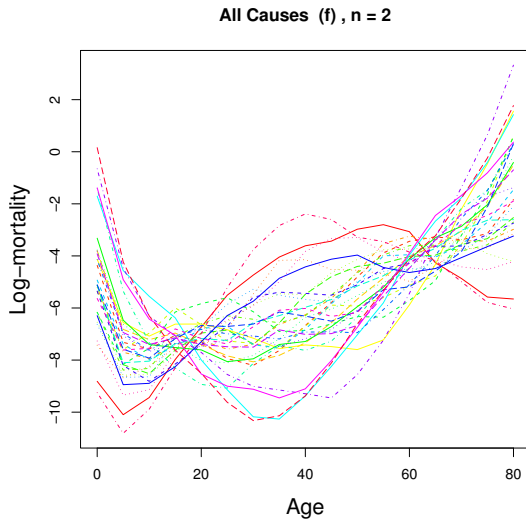
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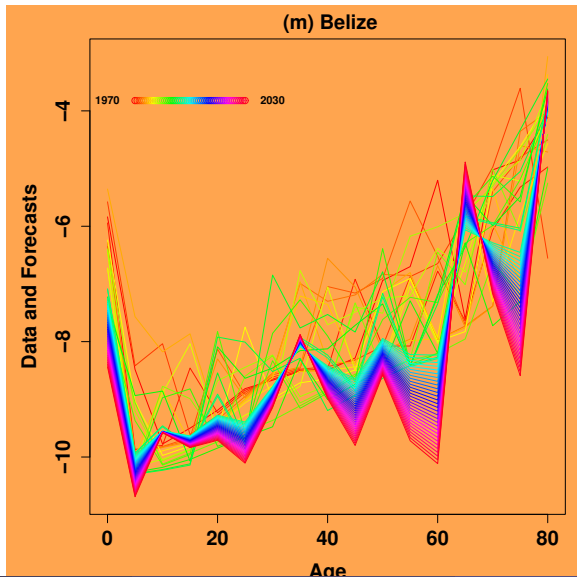
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- Can smooth simultaneously over different types of variables (age, country, and time).
- We can smooth interactions:
 - Smoothing *trends* over age groups.
 - Smoothing trends over age groups as they vary across countries, etc.
- The mathematical form for *all* these (separately or together) turns out to be the same:

$$\mathcal{P}(\boldsymbol{\beta} \mid \theta) \propto \exp \left(-\frac{\theta}{2} \sum_{ij} W_{ij} \boldsymbol{\beta}_i' \mathbf{C}_{ij} \boldsymbol{\beta}_j \right), \quad \mathbf{C}_{aa'} \equiv \frac{1}{T} \mathbf{Z}_a \mathbf{Z}_{a'}$$

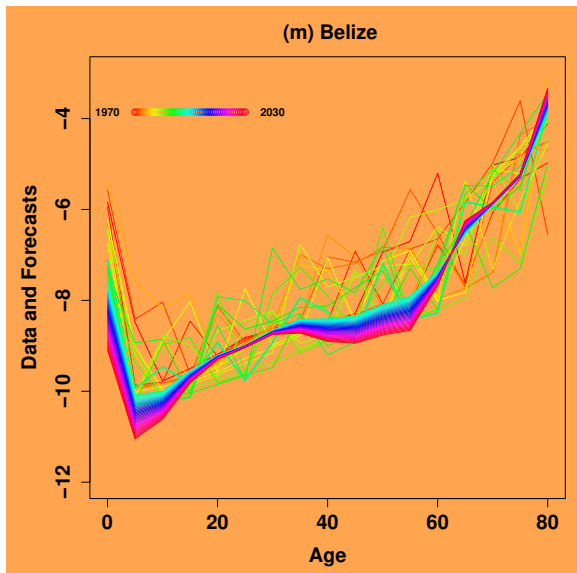
Mortality from Respiratory Infections, Males

Least Squares



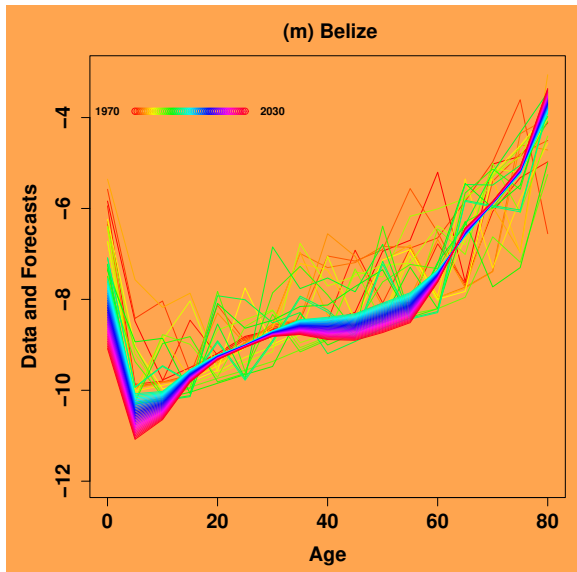
Mortality from Respiratory Infections, males, $\sigma = 2.00$

Smoothing over Age Groups



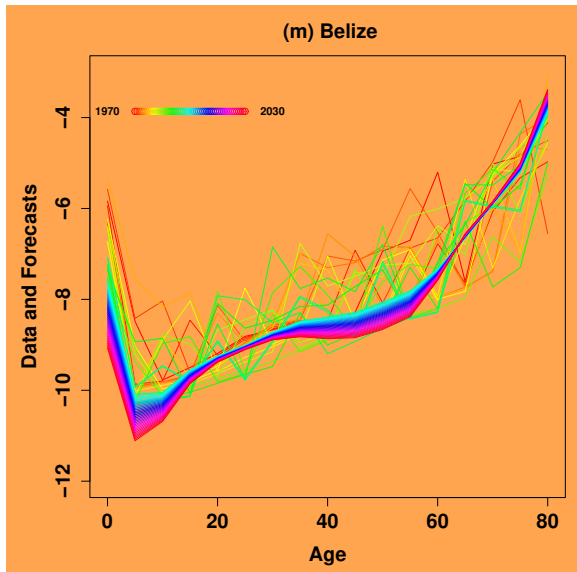
Mortality from Respiratory Infections, males, $\sigma = 1.51$

Smoothing over Age Groups



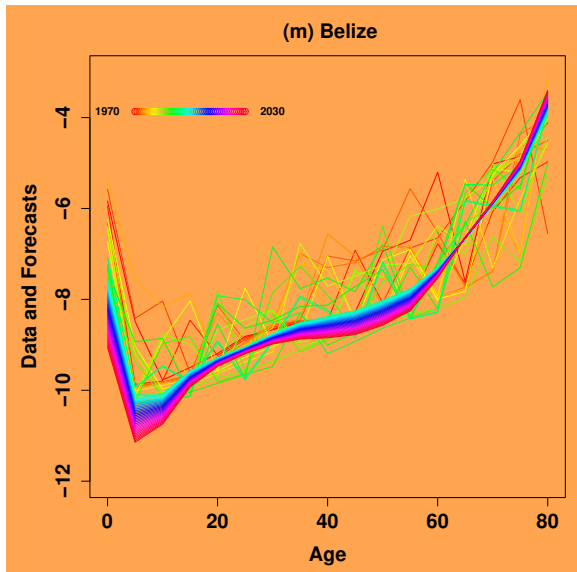
Mortality from Respiratory Infections, males, $\sigma = 1.15$

Smoothing over Age Groups



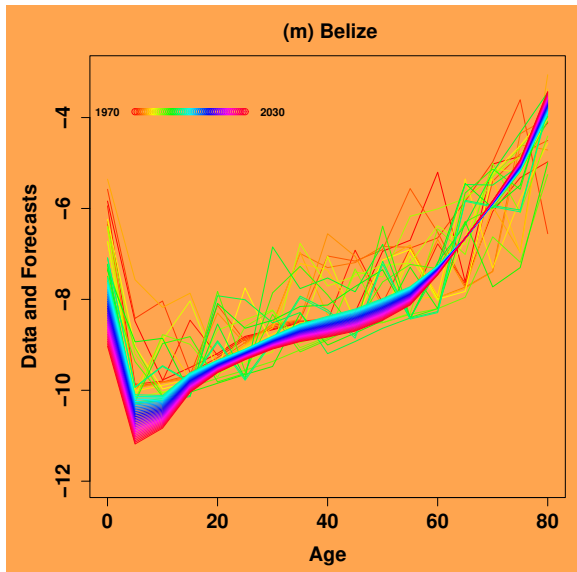
Mortality from Respiratory Infections, males, $\sigma = 0.87$

Smoothing over Age Groups



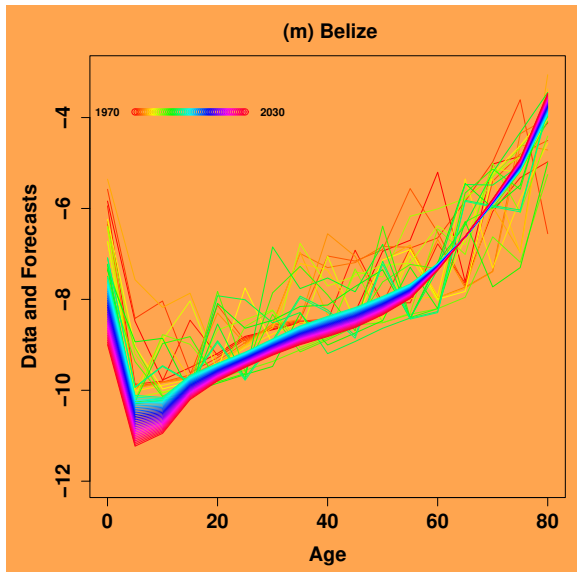
Mortality from Respiratory Infections, males, $\sigma = 0.66$

Smoothing over Age Groups



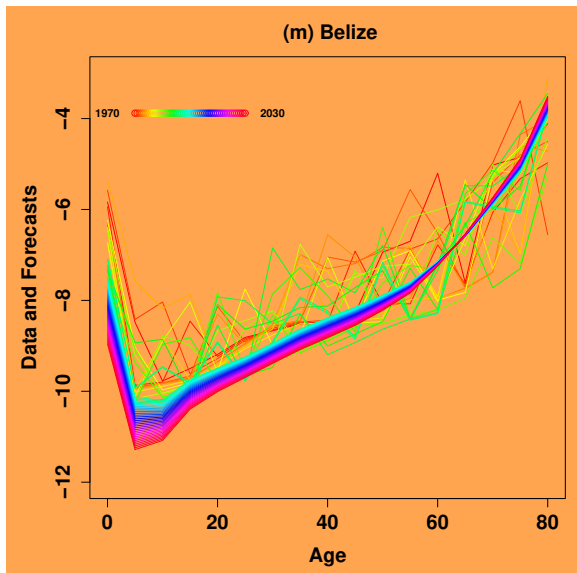
Mortality from Respiratory Infections, males, $\sigma = 0.50$

Smoothing over Age Groups



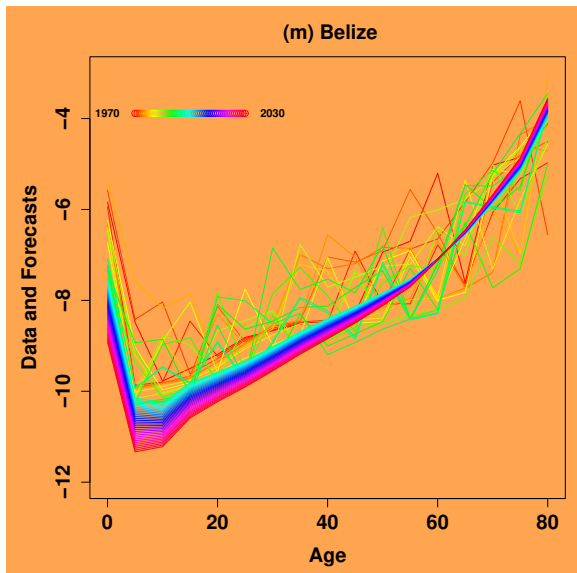
Mortality from Respiratory Infections, males, $\sigma = 0.38$

Smoothing over Age Groups



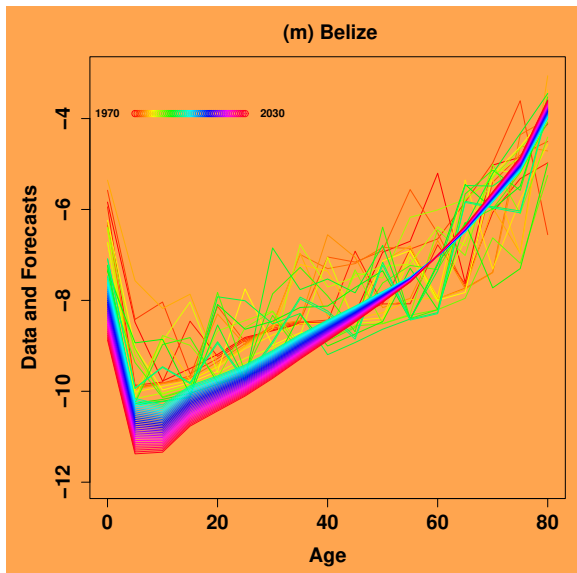
Mortality from Respiratory Infections, males, $\sigma = 0.28$

Smoothing over Age Groups



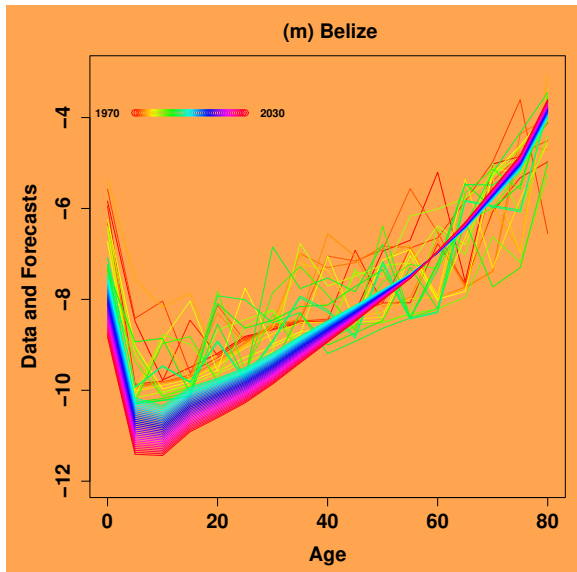
Mortality from Respiratory Infections, males, $\sigma = 0.21$

Smoothing over Age Groups



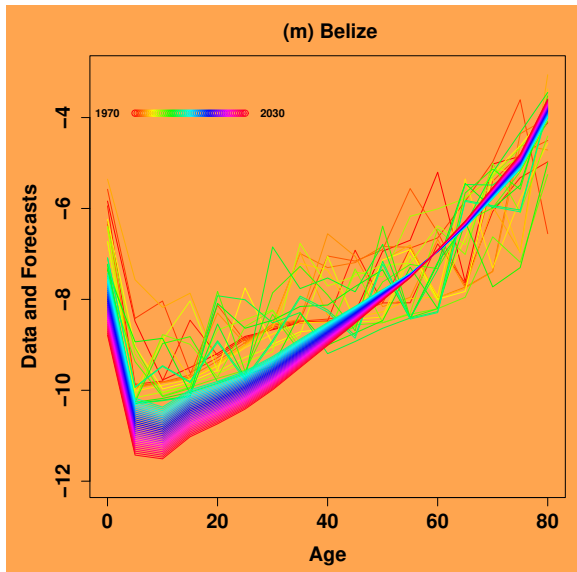
Mortality from Respiratory Infections, males, $\sigma = 0.16$

Smoothing over Age Groups



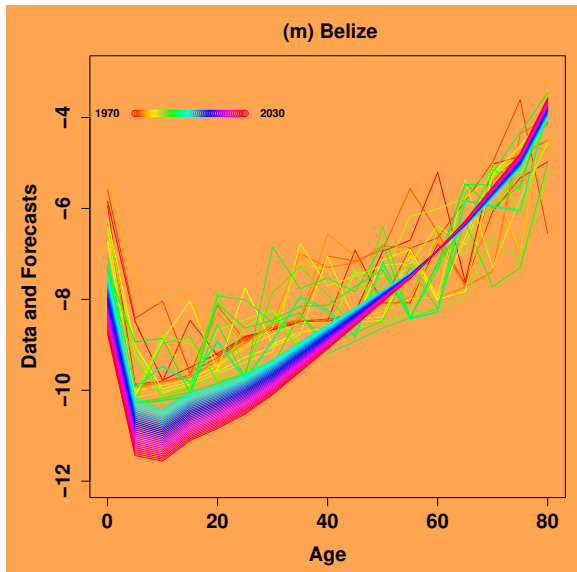
Mortality from Respiratory Infections, males, $\sigma = 0.12$

Smoothing over Age Groups



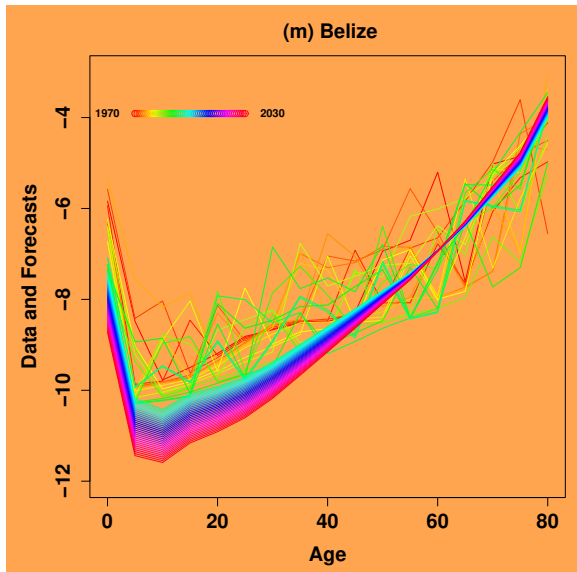
Mortality from Respiratory Infections, males, $\sigma = 0.09$

Smoothing over Age Groups



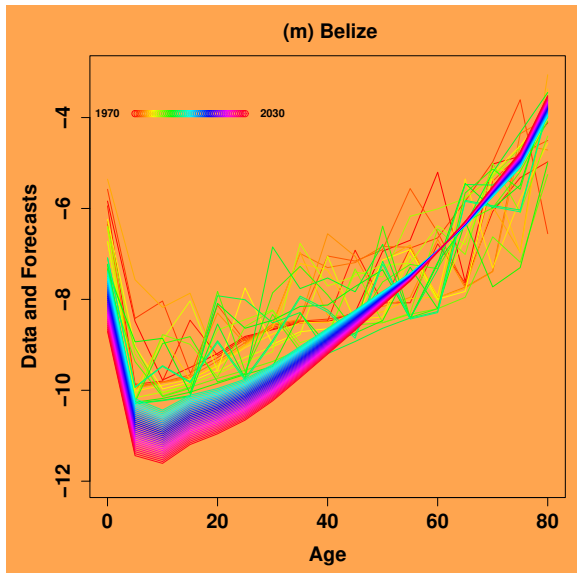
Mortality from Respiratory Infections, males, $\sigma = 0.07$

Smoothing over Age Groups



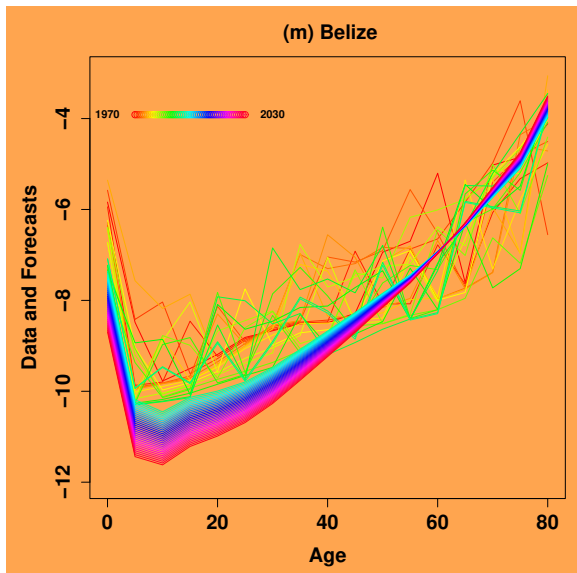
Mortality from Respiratory Infections, males, $\sigma = 0.05$

Smoothing over Age Groups



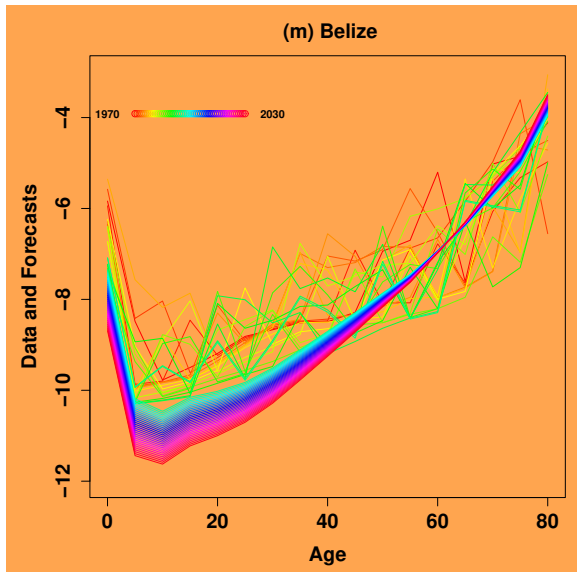
Mortality from Respiratory Infections, males, $\sigma = 0.04$

Smoothing over Age Groups



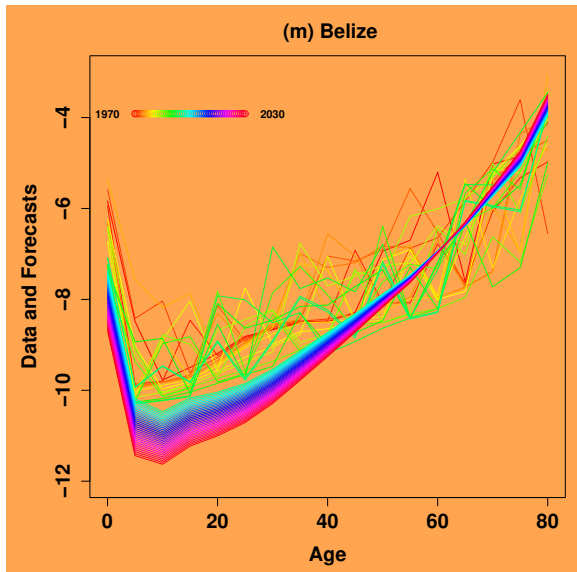
Mortality from Respiratory Infections, males, $\sigma = 0.03$

Smoothing over Age Groups



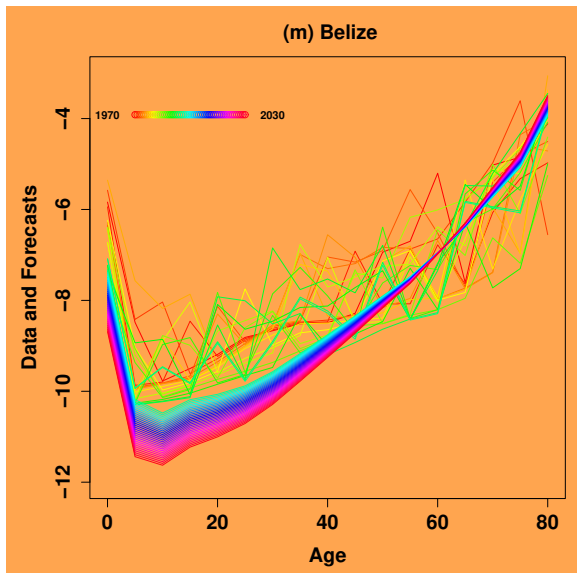
Mortality from Respiratory Infections, males, $\sigma = 0.02$

Smoothing over Age Groups



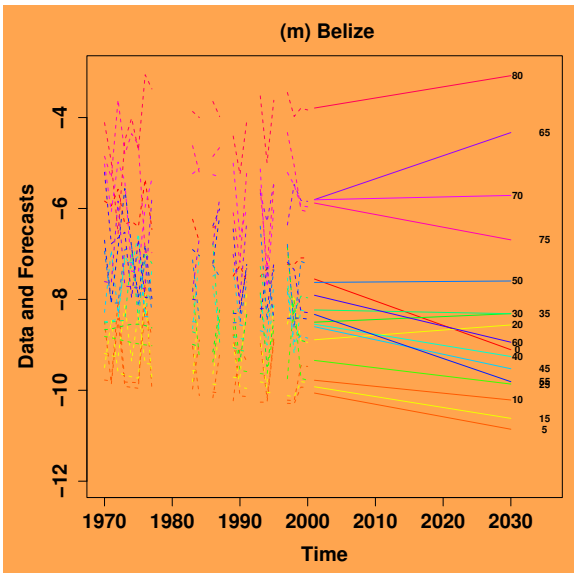
Mortality from Respiratory Infections, males, $\sigma = 0.01$

Smoothing over Age Groups



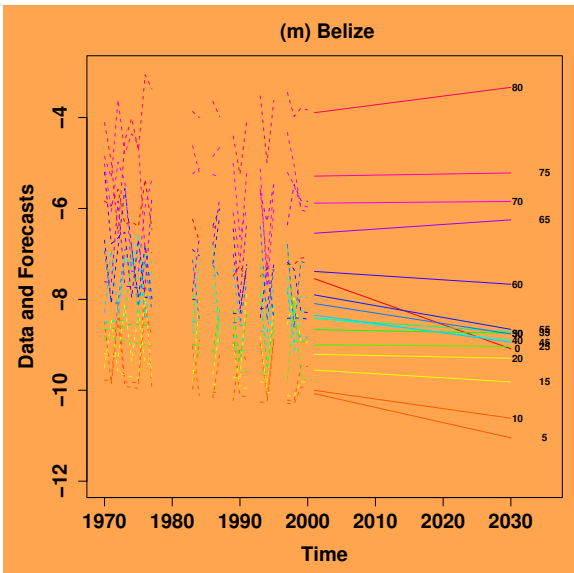
Mortality from Respiratory Infections, males

Least Squares



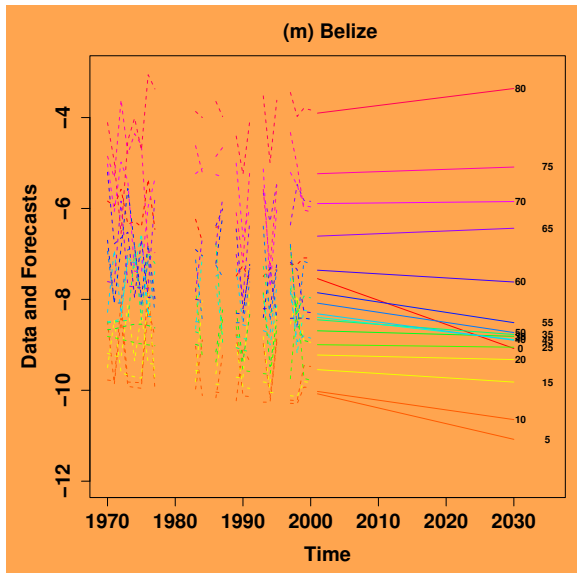
Mortality from Respiratory Infections, males, $\sigma = 2.00$

Smoothing over Age Groups



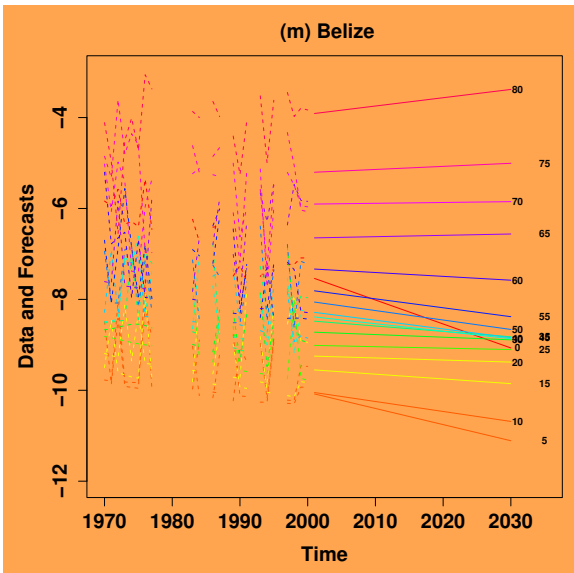
Mortality from Respiratory Infections, males, $\sigma = 1.51$

Smoothing over Age Groups



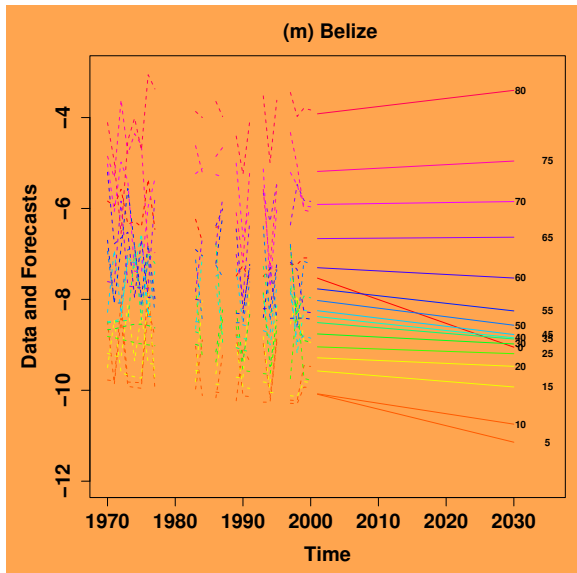
Mortality from Respiratory Infections, males, $\sigma = 1.15$

Smoothing over Age Groups



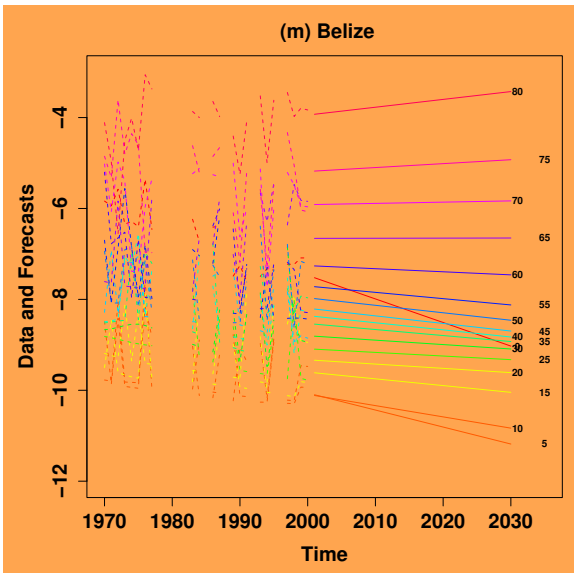
Mortality from Respiratory Infections, males, $\sigma = 0.87$

Smoothing over Age Groups



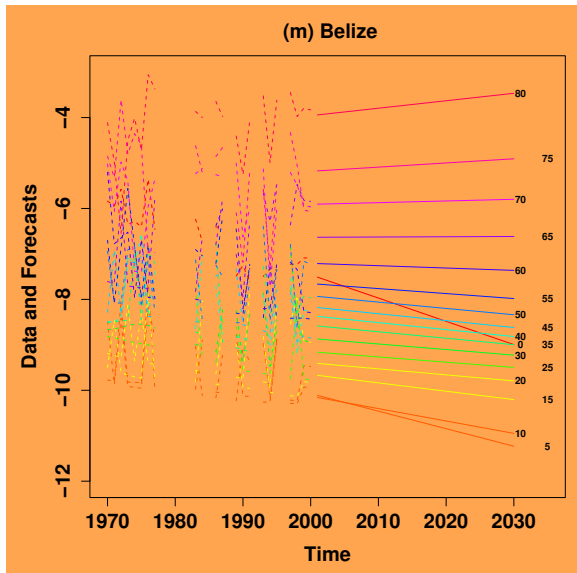
Mortality from Respiratory Infections, males, $\sigma = 0.66$

Smoothing over Age Groups



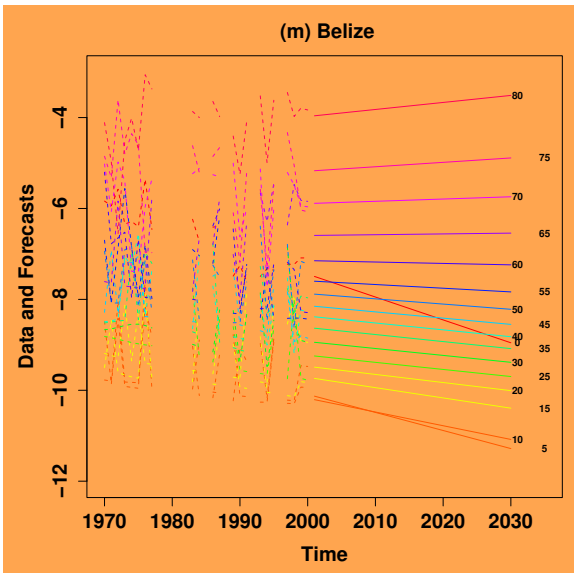
Mortality from Respiratory Infections, males, $\sigma = 0.50$

Smoothing over Age Groups



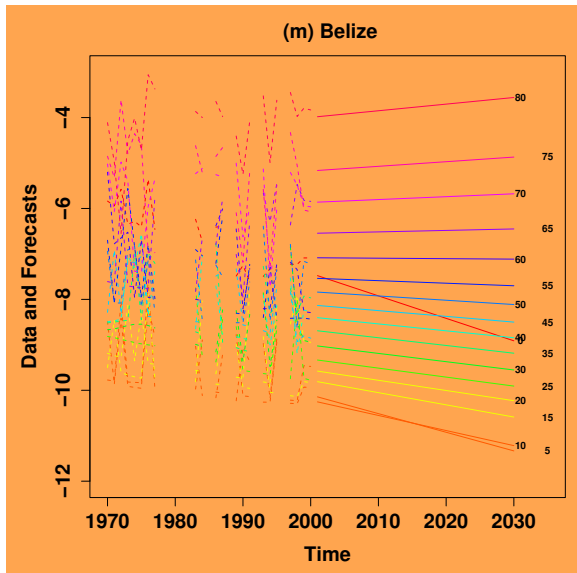
Mortality from Respiratory Infections, males, $\sigma = 0.38$

Smoothing over Age Groups



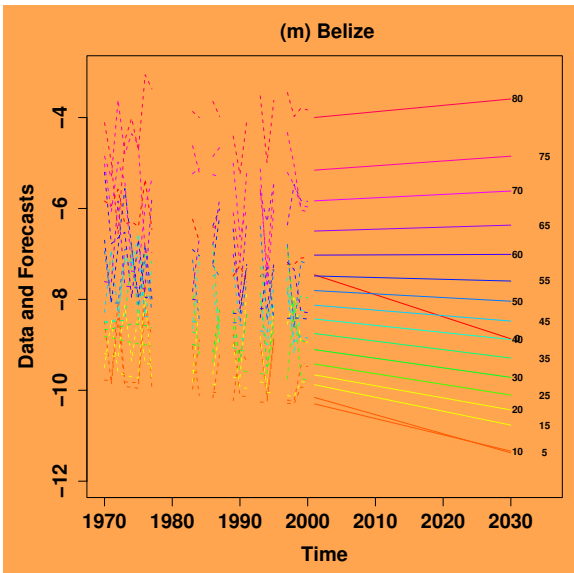
Mortality from Respiratory Infections, males, $\sigma = 0.28$

Smoothing over Age Groups



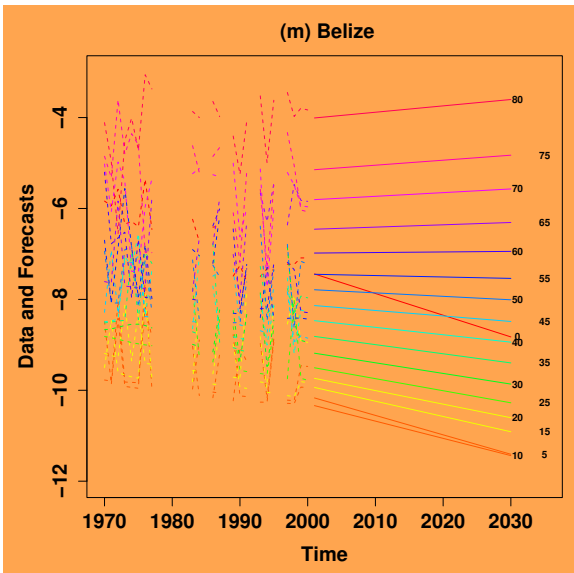
Mortality from Respiratory Infections, males, $\sigma = 0.21$

Smoothing over Age Groups



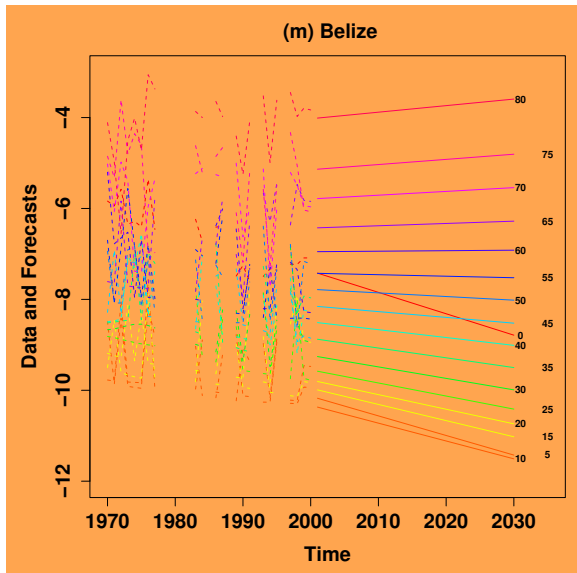
Mortality from Respiratory Infections, males, $\sigma = 0.16$

Smoothing over Age Groups



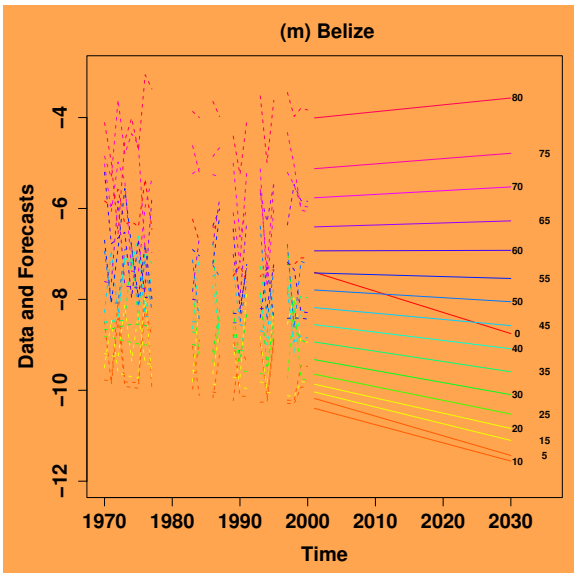
Mortality from Respiratory Infections, males, $\sigma = 0.12$

Smoothing over Age Groups



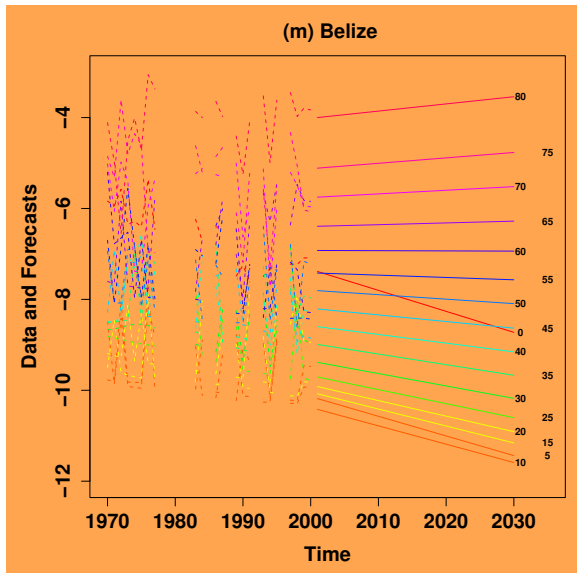
Mortality from Respiratory Infections, males, $\sigma = 0.09$

Smoothing over Age Groups



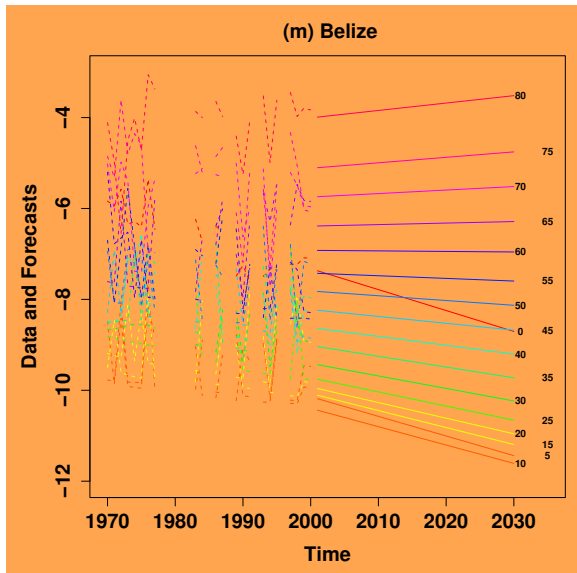
Mortality from Respiratory Infections, males, $\sigma = 0.07$

Smoothing over Age Groups



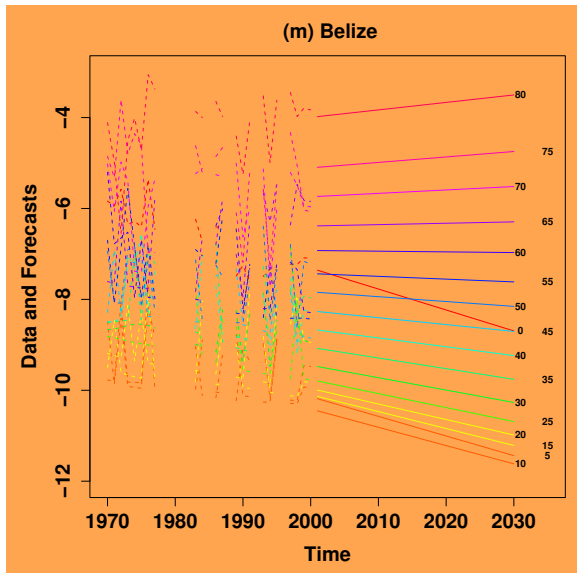
Mortality from Respiratory Infections, males, $\sigma = 0.05$

Smoothing over Age Groups



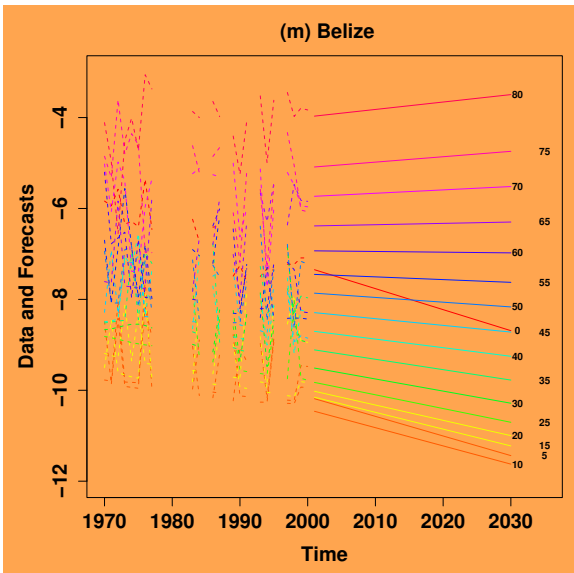
Mortality from Respiratory Infections, males, $\sigma = 0.04$

Smoothing over Age Groups



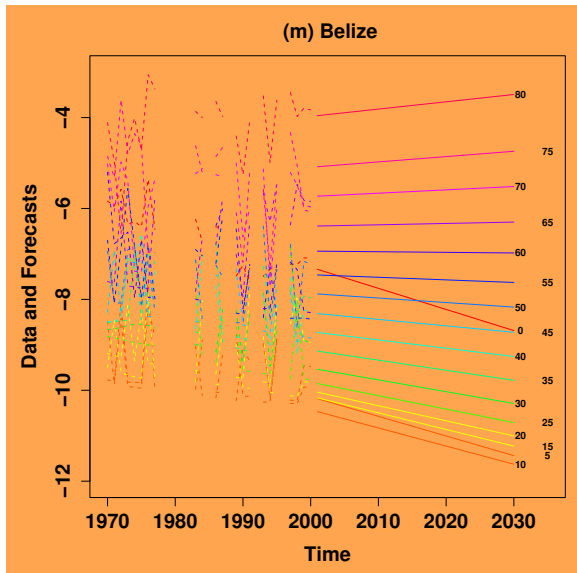
Mortality from Respiratory Infections, males, $\sigma = 0.03$

Smoothing over Age Groups



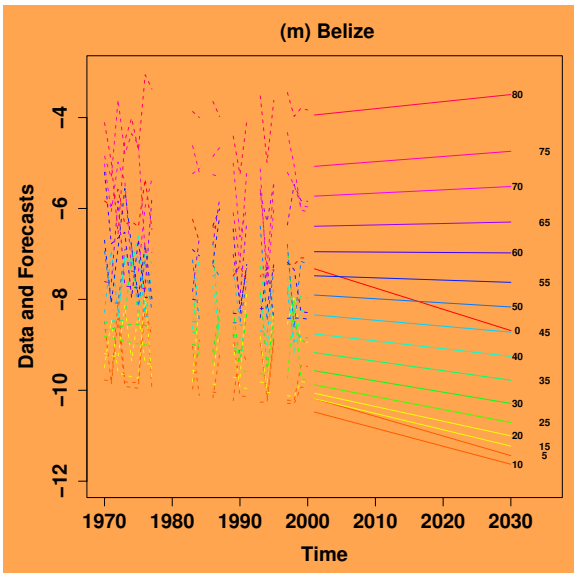
Mortality from Respiratory Infections, males, $\sigma = 0.02$

Smoothing over Age Groups



Mortality from Respiratory Infections, males, $\sigma = 0.01$

Smoothing over Age Groups



Smoothing Trends over Age Groups

Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Smoothing Trends over Age Groups

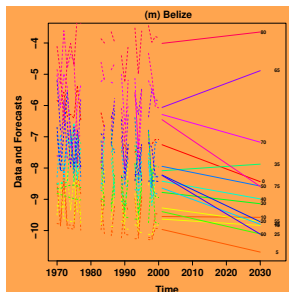
Log-mortality in Belize males from respiratory infections

Least Squares

Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

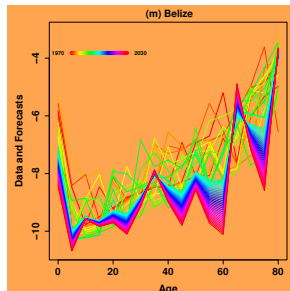
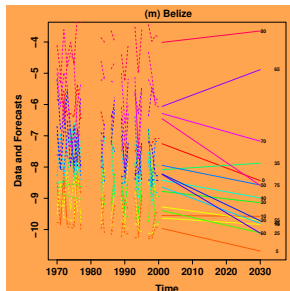
Least Squares



Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

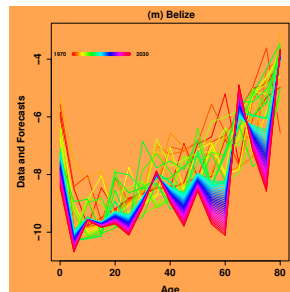
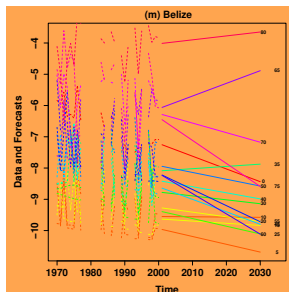
Least Squares



Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Least Squares

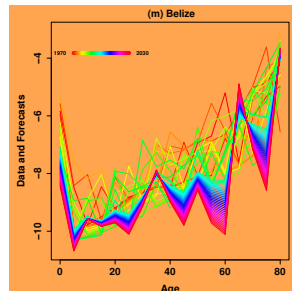
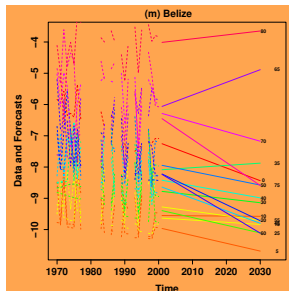


Smoothing
Age Groups

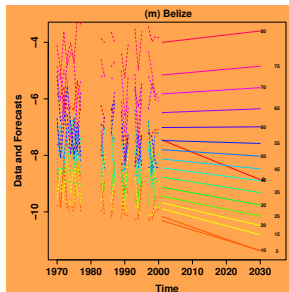
Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Least Squares



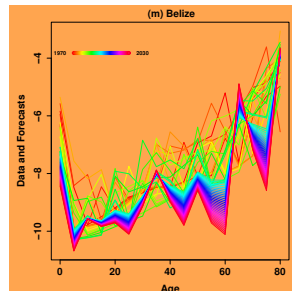
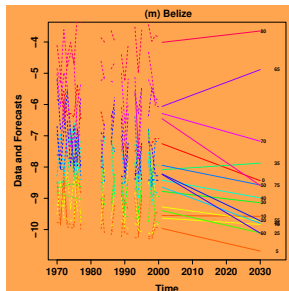
Smoothing
Age Groups



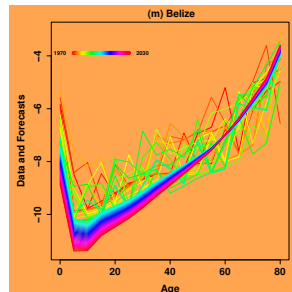
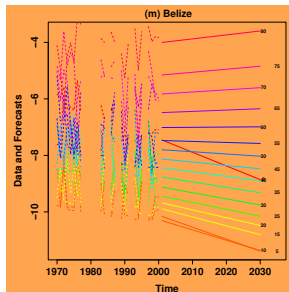
Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Least Squares



Smoothing
Age Groups



Smoothing Trends over Age Groups and Time

Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

Smoothing Trends over Age Groups and Time

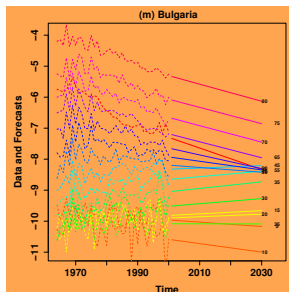
Log-Mortality in Bulgarian males from respiratory infections

Least Squares

Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

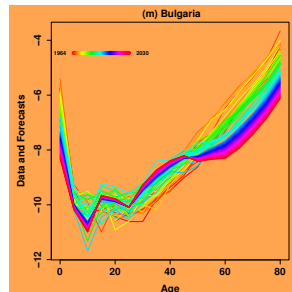
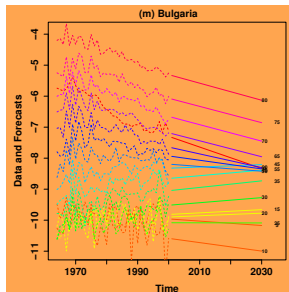
Least Squares



Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

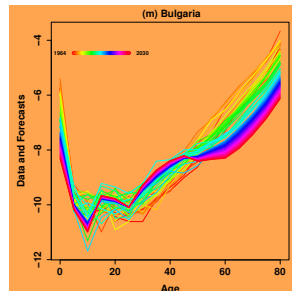
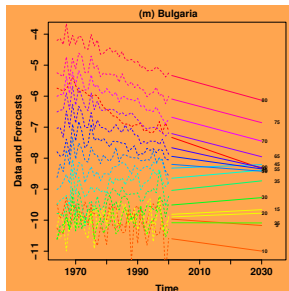
Least Squares



Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

Least Squares

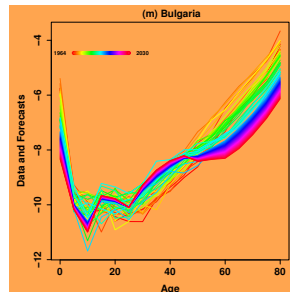
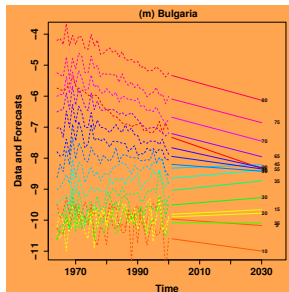


Smoothing
Age and Time

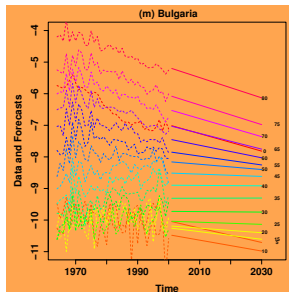
Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

Least Squares



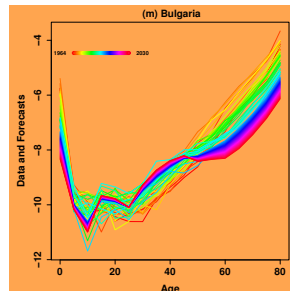
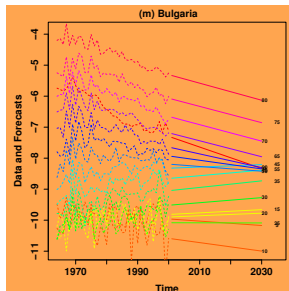
Smoothing
Age and Time



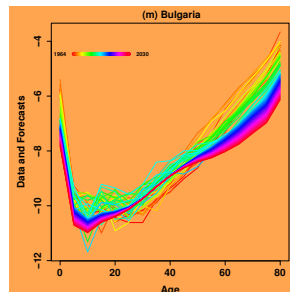
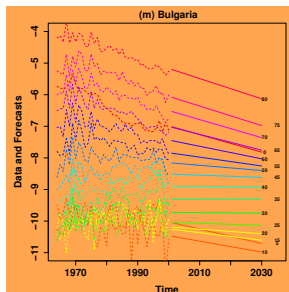
Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

Least Squares



Smoothing
Age and Time



Using Covariates (GDP, tobacco, trend, log trend)

Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

Using Covariates (GDP, tobacco, trend, log trend)

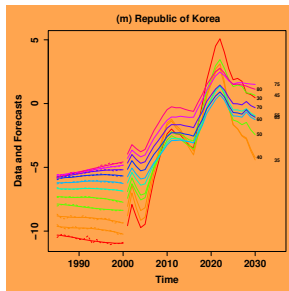
Lung cancer in Korean Males

Least Squares

Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

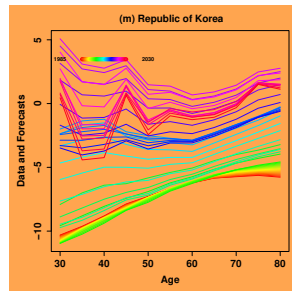
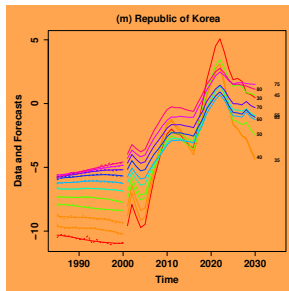
Least Squares



Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

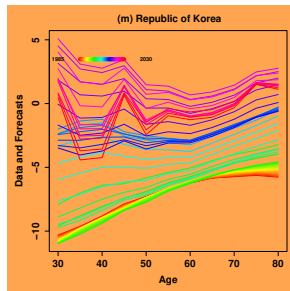
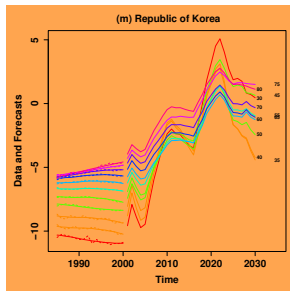
Least Squares



Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

Least Squares

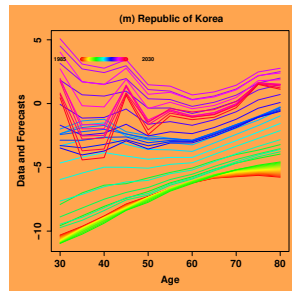
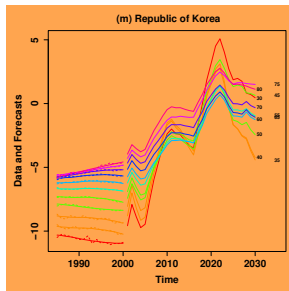


Smooth over age,
time, age/time

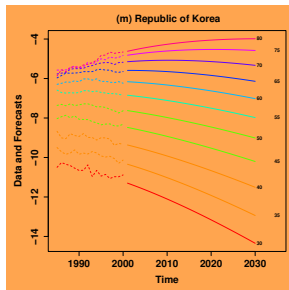
Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

Least Squares



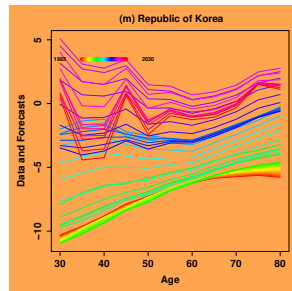
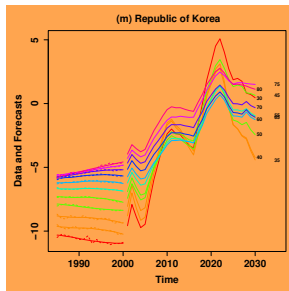
Smooth over age,
time, age/time



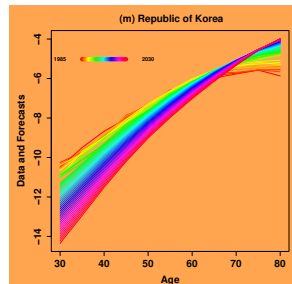
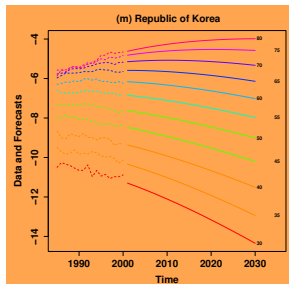
Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

Least Squares



Smooth over age,
time, age/time



Using Covariates (GDP, tobacco, trend, log trend)

Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Males, Singapore

Using Covariates (GDP, tobacco, trend, log trend)

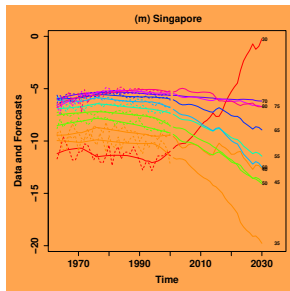
Lung cancer in Males, Singapore

Least Squares

Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Males, Singapore

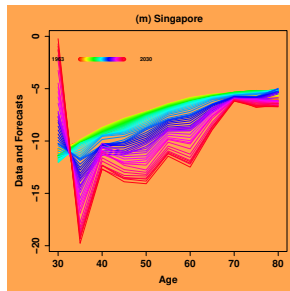
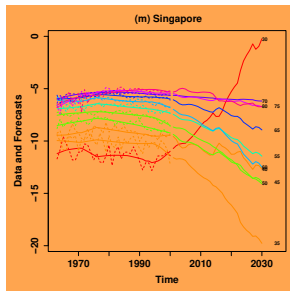
Least Squares



Using Covariates (GDP, tobacco, trend, log trend)

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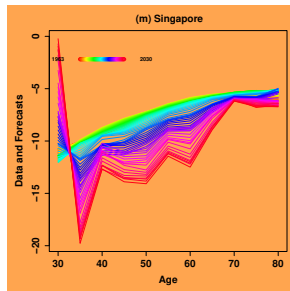
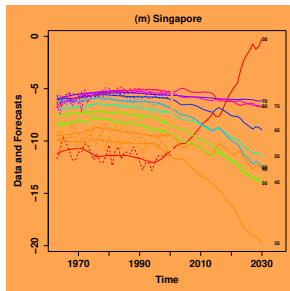
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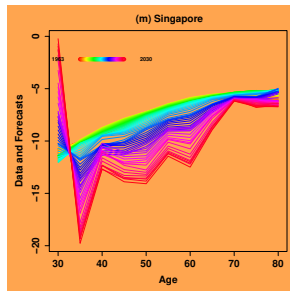
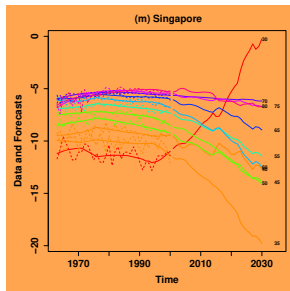


Smooth over age,
time, age/time

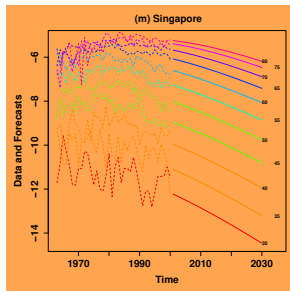
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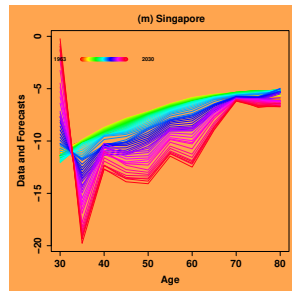
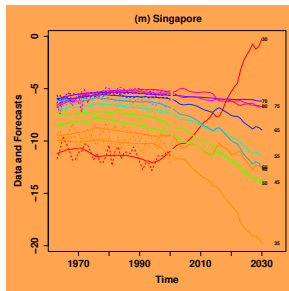
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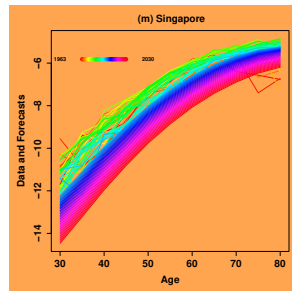
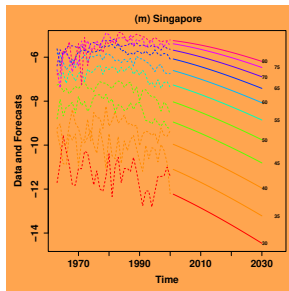
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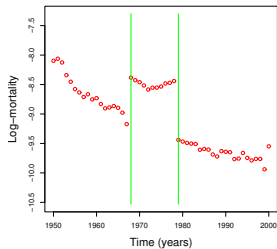


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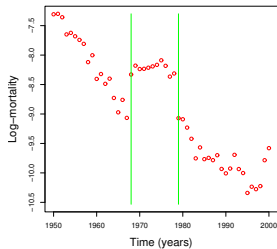


What about ICD Changes?

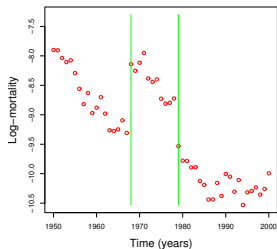
Other Infectious Diseases : USA , age 0 (m)



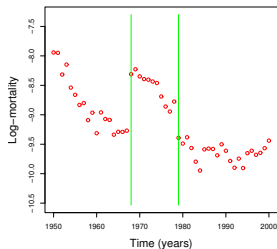
Other Infectious Diseases : France , age 0 (m)



Other Infectious Diseases : Australia , age 0 (m)

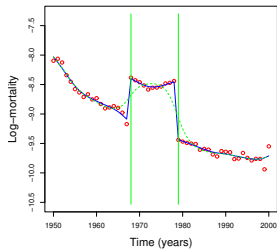


Other Infectious Diseases : United Kingdom , age 0 (m)

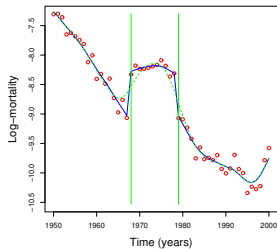


Fixing ICD Changes

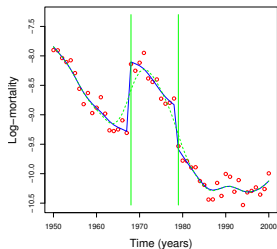
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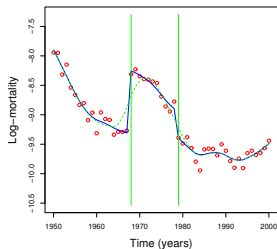
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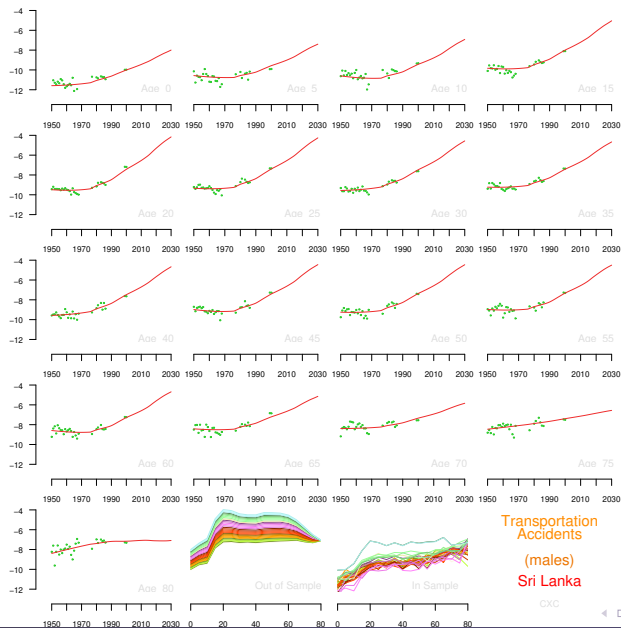
Other Infectious Diseases : United Kingdom , age 0 (m)



A book manuscript, YourCast software, etc.

<http://GKing.Harvard.edu>

Without Country Smoothing

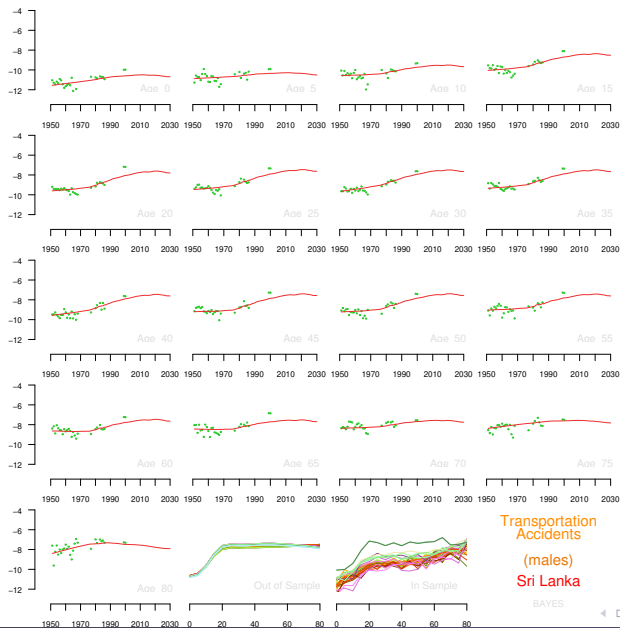


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Demographic Forecasting

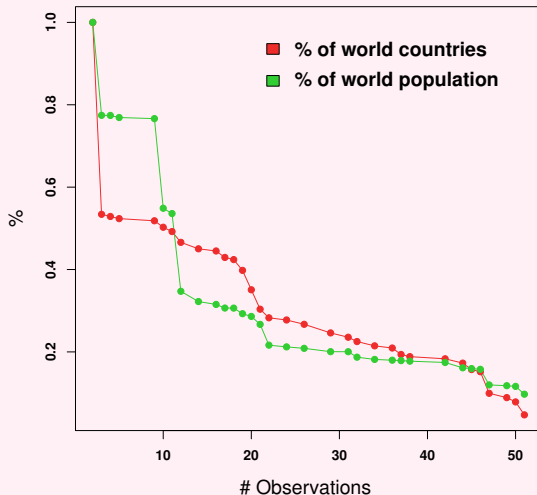
91 / 96

With Country Smoothing



Many Short Time Series

Coverage of WHO data base (age specific, all causes)



Prior Indifference

- These priors are “indifferent” to transformations:

$$\mu(a, t) \rightsquigarrow \mu(a, t) + p(a, t)$$

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- where $p(a, t)$ is a polynomial in a (whose degree is the degree of the derivative in the prior)
- Prior information is about **relative** (not absolute) levels of log-mortality

Preview of Results: Out-of-Sample Evaluation

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Mean Absolute Error in Males (over age and country)

Preview of Results: Out-of-Sample Evaluation

Mean Absolute Error in Males (over age and country)

	<u>% Improvement</u>	
	Over Best Previous	to Best Conceivable
Cardiovascular	22	49
Lung Cancer	24	47
Transportation	16	31
Respiratory Chronic	13	30
Other Infectious	12	30
Stomach Cancer	8	24
All-Cause	12	22
Suicide	7	17
Respiratory Infectious	3	7

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- Each row averages 6,800 forecast errors (17 age groups, 40 countries, and 10 out-of-sample years).

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- Does *considerably* better with **more informative covariates**

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Mean Absolute Error in Males (over age and country)

	Mean Absolute Error			% Improvement	
	Best Previous	Our Method	Best Conceivable	Over Best Previous	to Best Conceivable
Cardiovascular	0.34	0.27	0.19	22	49
Lung Cancer	0.36	0.27	0.17	24	47
Transportation	0.37	0.31	0.18	16	31
Respiratory Chronic	0.45	0.39	0.26	13	30
Other Infectious	0.55	0.48	0.32	12	30
Stomach Cancer	0.30	0.27	0.20	8	24
All-Cause	0.17	0.15	0.08	12	22
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- Does much better with better covariates